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AI Super-Resolution of Satellite Imagery: The Evidential Paradigm Shift in CAP Monitoring

Technical and Operational Considerations

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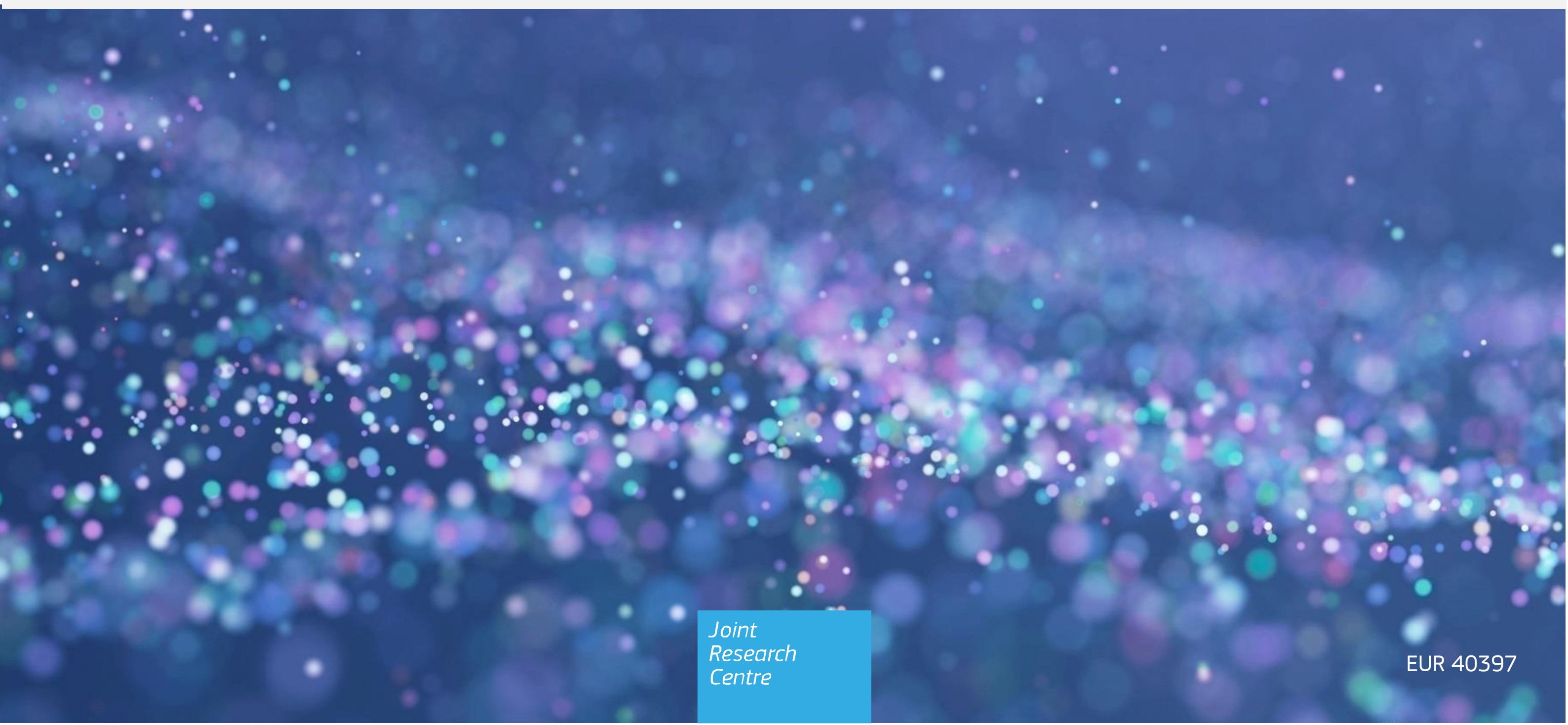

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Contents

Abstract

This report examines AI-based super-resolution (AI-SR) technology within Common Agricultural Policy (CAP) monitoring systems, analysing its paradigm shift in satellite-based evidence. This technology represents a fundamental change, as satellite imagery evolves from direct documentation of physical reality to enhanced representations that may incorporate visually convincing but non-existent features. This transformation creates specific considerations regarding evidential reliability and regulatory compliance. The report proposes a certification framework to ensure compliance with Articles 10(3) and 11 of Implementing Regulation (EU) 2022/1173, which stipulate that alternative data sources must offer at least equivalent value for the purpose intended and ensure sufficient assurance on the accuracy of the area declared and on compliance with the relevant obligations. Through examination of technical foundations, performance boundaries, operational impacts, and deployment strategies, this analysis provides Member States' paying agencies with a structured evaluation framework. This guidance aims to facilitate evidence-based decision-making, ensuring that AI-enhanced imagery can be responsibly and reliably integrated into CAP monitoring systems.

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We extend particular appreciation to DigiFarm AS for providing examples of AI-based super-resolution imagery for this technical report. Their contribution reflects a commitment to transparent technical discourse and the collaborative advancement of agricultural monitoring technologies. Providing operational examples that illustrate both capabilities and current constraints shows commendable openness, which advances collective understanding within the agricultural monitoring community.

We also thank the SR4RS framework developers for the publicly available imagery examples used from their documentation.

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Executive summary

As CAP monitoring systems increasingly rely on satellite-based remote sensing for efficient, data-driven agricultural monitoring, the emergence of AI-based super-resolution (AI-SR) technology offers new opportunities by artificially enhancing spatial resolution, potentially addressing Sentinel-2's main limitation, leading to expectations of improved monitoring capabilities.

While promising, this technology involves a fundamental trade-off between enhanced spatial resolution and evidentiary data integrity. This constitutes a paradigm shift that transforms the evidential trustworthiness of satellite imagery.

Traditional satellite imagery has been inherently trusted as evidence because it provides direct capture of ground conditions through well-understood optical and electronic processes.

In contrast, AI-SR generates partially synthetic imagery through statistical inference. Specifically, AI-SR does not recover "hidden" details from low-resolution images; instead, it makes educated guesses to create details based on what it learned from thousands of paired low and high-resolution satellite images. Unfortunately, this process can inadvertently introduce non-existent features or eliminate/replace actual features, producing imagery that appears visually convincing but may not accurately represent actual ground conditions.

Given these evidential challenges, this report establishes a systematic evaluation framework for assessing AI-SR technology within CAP monitoring systems. Recognising that this paradigm shift necessitates analysis of evidential value implications, the framework provides methodologies for evaluating technical constraints, evidential requirements, and operational considerations, enabling paying agencies to make informed decisions.

The analysis examines four critical dimensions: (1) technical foundations of AI-SR, including its evolution to advanced deep learning architectures and inherent limitations illustrated through operational examples; (2) performance boundaries that establish relationships between enhancement factors and output reliability; (3) operational impacts on key monitoring functions, from parcel boundary update to agricultural practice verification; and (4) certification frameworks to establish the evidentiary value of enhanced imagery.

Within this analytical context, Implementing Regulation (EU) 2022/1173 permits Member States to utilize alternative data sources provided they "offer at least equivalent value for the purpose intended" (Article 10(3)) while ensuring "sufficient assurance on the accuracy of the area declared and on compliance with the relevant obligations" (Article 11). Data quality adequacy is determined through Member States' established quality assurance processes. The complete regulatory framework is available through the EUR-Lex portal (https://eur-lex.europa.eu/eli/reg_impl/2022/1173/oj/eng)

To support this systematic evaluation process, this report employs a tiered information architecture designed for different purposes and levels of detail. Paying agency decision-makers can focus on the main body (Sections 1-7) for strategic analysis of capabilities, limitations, and regulatory implications essential for policy and implementation decisions. Technical experts can access technical details and methodological frameworks through the systematically organized Annexes (1-5), which are referenced throughout the main text. This approach ensures paying agencies can efficiently access information appropriate to their specific needs and responsibilities.

1 Introduction: The Evolution of Agricultural Monitoring

The implementation of the Common Agricultural Policy (CAP) has undergone a fundamental transformation through the integration of satellite-based remote sensing and complementary technologies, shifting from traditional spot checks to systematic, data-driven assessment of agricultural practices and compliance. This evolution demands efficient, accurate, and cost-effective monitoring systems capable of operating at large scales. As monitoring becomes more sophisticated, advances in satellite imaging and artificial intelligence present both opportunities and challenges. Within this context, paying agencies may wonder whether AI-based super-resolution (AI-SR) technology might address resolution limitations in freely available satellite imagery, particularly from the Sentinel-2 platform.

To address these considerations systematically, this report examines whether and how AI-SR technology could enhance CAP monitoring frameworks, while considering technical constraints, implementation challenges, and quality assurance considerations. The analysis focuses on three fundamental questions:

— How might AI super-resolution technology enhance the utility of Sentinel-2 imagery within CAP's monitoring system while ensuring reliability in an operational framework ?

— What are the inherent limitations and potential risks of this technology in agricultural monitoring applications, and how can these risks be mitigated ?

— What certification and quality assurance frameworks would be appropriate to establish the evidential value of AI-enhanced imagery?

These questions guide the report's analytical structure:

— **Technical Foundations** (Section 2): Analyses AI super-resolution technology's development, tracing its evolution from early computational approaches to advanced deep learning architectures while identifying characteristic constraints that shape implementation possibilities.

— **Fundamental Constraints and Technical Boundaries** (Section 3): Establishes a systematic framework for understanding AI-SR limitations, addressing the Sentinel-2 platform's characteristics and the relationship between scaling factors and monitoring reliability.

— **Implementation Considerations** (Section 4) : Provides paying agencies with evaluation frameworks to inform decision-making processes, with technical details of both open-source frameworks and commercial solutions provided in Annexes 3.2 and 3.3.

— **Operational Impacts** (Section 5) : Assesses how AI-SR's technical limitations specifically affect key CAP monitoring functions - from parcel boundary update to landscape feature identification and agricultural practice monitoring.

— **Evidential Value and Certification** (Section 6): Proposes a certification framework with structured artifact detection methodologies, validation protocols, and quality standards necessary to reestablish enhanced imagery's evidentiary value.

Through this structured evaluation approach, paying agencies can assess AI-SR within their specific operational contexts while maintaining the reliability and evidential value essential for compliance - supporting informed technology decisions rather than prescribing specific implementation pathways.

2 Technical Foundations: Evolution and State-of-the-Art in AI Super-Resolution Technology

2.1 Conceptual Framework

As Earth observation applications increasingly demand detailed spatial information, the ability to enhance image resolution has become more desirable. Advanced technologies like super-resolution (SR) may - up to a certain point - enable this enhancement. Understanding how these technologies actually function is essential for evaluating their potential role within the CAP monitoring frameworks.

2.1.1 Understanding AI Super-Resolution: Technical Reality vs. Common Perception

AI-based super-resolution represents a domain where popular understanding may diverge from the underlying technical reality. This section addresses fundamental conceptual assumptions to establish an appropriate analytical framework for subsequent technical assessment.

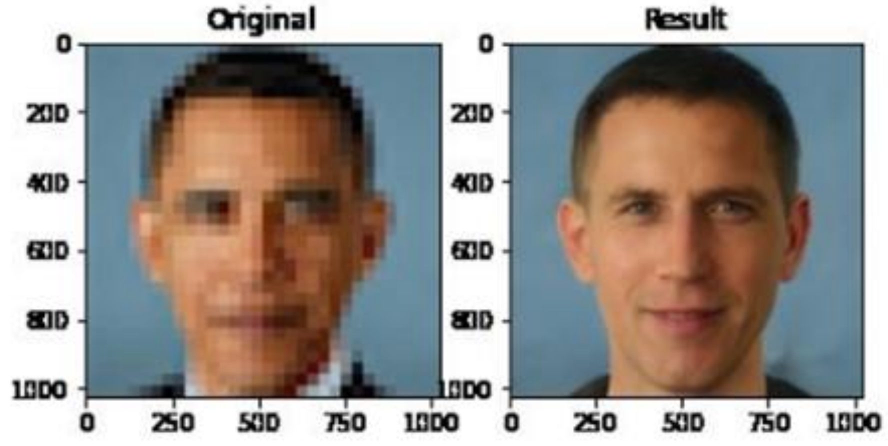
Low-resolution imagery fundamentally lacks detailed spatial information - a critical technical constraint that underlies all AI-SR applications. The resolution of an acquisition device (camera or satellite sensor) directly determines the level of detail captured during image acquisition. When imagery is acquired at low resolution, fine details below the sensor's resolving capability are not recorded - they are not merely "hidden" but absent from the digital representation. This information loss during acquisition represents an irreversible process, rendering it physically impossible to recover details that were never captured by the sensing apparatus.

Despite this fundamental physical constraint, a persistent misconception regarding AI-SR technology suggests that it extracts or uncovers details that exist but remain imperceptible in low-resolution imagery. This interpretation fundamentally mischaracterizes the operational principles of AI-SR systems.

Rather than extracting or recovering "hidden" details, an AI-SR model uses statistical inference to generate plausible missing details based on patterns learned during training. In essence, the system performs a form of "informed guessing" - it makes educated predictions about what high-resolution details might look like based on patterns of correlation learned from thousands of paired datasets of low and high-resolution training images.

Figure 1 demonstrates this principle through a compelling empirical example. The low-resolution image on the left depicts former U.S. President Barack Obama, recognizable despite significant pixelation. However, the AI-enhanced image on the right, which is the output of the super-resolution process, reveals an entirely different individual with facial features that, while realistic and well-defined, do not represent Obama or any actual person.

Figure 1. Comparison of a low-resolution image of Barack Obama (left) with its AI super-resolution output (right), generated using PULSE, a GAN-based super-resolution model, demonstrating how AI-SR generates new features rather than recovering actual information.



Source: X.com, © *@Chicken3gg, (June 20, 2020)*

This transformation conclusively illustrates that AI-SR generates synthetic content through pattern-based inference rather than recovering actual information. The enhanced output represents a statistical prediction based on limited input data rather than an accurate representation of how Obama would appear if photographed with a high-resolution camera. While more advanced AI-SR models trained on larger datasets might produce more visually convincing results, the fundamental principle remains unchanged: all enhanced details are synthetically generated, not recovered from the original image. This fundamentally challenges the misinterpretation that AI-SR outputs constitute "true" high-resolution representations of the original scene.

The underlying principle applies universally across AI-SR implementations, whether for portrait photography or satellite imagery: these systems analyse low-resolution data to identify basic structures, then generate plausible details based on training patterns. The algorithm produces synthetic details through statistical inference, not by revealing hidden information.

Understanding the distinction between information recovery and synthetic generation is essential for proper implementation of AI-SR within the CAP monitoring frameworks, particularly for applications where the distinction between actual and synthetically generated details may have significant implications for eligibility assessment or compliance monitoring.

These fundamental principles of statistical inference manifest across various operational contexts, from infrastructure recognition challenges (Section 3.1, Figure 2) to boundary precision issues in parcel delineation (Section 5.1, Figure 3) and small landscape feature preservation (Section 5.2, Figure 4).

2.1.2 Evidential Implications: The Paradigm Shift in Regulatory Evidence

Image-based evidence, including satellite imagery, aerial photography, and ground-based photography, has historically formed a cornerstone of regulatory and evidential frameworks. Traditional imagery content has historically been inherently trusted to accurately represent ground reality because it stems from direct capture of actual conditions through well-understood optical and electronic processes. This deterministic mechanism produces a systematic, physically traceable relationship between the photons received by the sensor and the pixel values recorded, preserving a continuous chain of causality from scene to sensor.

The emergence of AI-based enhancement technologies represents a fundamental paradigm shift in the nature of image-based evidence. Unlike traditional imagery that captures actual ground conditions, AI-SR generates partially synthetic content through statistical inference rather than direct observation. This transformation means that enhanced imagery cannot be assumed to maintain the inherent trustworthiness of traditional satellite data, as the enhancement process may introduce features that appear contextually plausible but do not constitute a rigorous representation of the actual ground reality. Consequently, the use of AI-enhanced imagery within regulatory frameworks requires establishing specific safeguards and validation protocols to ensure appropriate evidential value. This report provides technical guidance and certification frameworks to support paying agencies in addressing these evidential considerations and ensuring that AI-enhanced imagery meets the 'equivalent value' standard established in Implementing Regulation (EU) 2022/1173.

2.2 Technological Evolution and Deep Learning Approaches: Overview

The evolution of super-resolution technology has progressed from simple mathematical interpolation methods to sophisticated AI-driven approaches. This section provides a summary of key developments and current state-of-the-art techniques. For technical details on deep learning architectures used in AI-based super-resolution (AI-SR) technology, including SRCNN, VDSR, SRGAN, and specialized frameworks for satellite imagery, please refer to Annex 1.1: Technical Foundations: Deep Learning Architectures for Super-Resolution.

Early super-resolution techniques used mathematical interpolation methods to predict what new pixels should look like when expanding a low-resolution image to a larger size, applying mathematical functions (like cubic polynomials) to nearby pixels in the original version. These methods essentially 'filled in the gaps' by averaging neighbouring pixel values, producing blurry, unrealistic images that failed to capture fine details or preserve the spectral characteristics essential for satellite imagery analysis.

The field underwent significant transformation with machine learning and deep learning approaches, which enabled the pattern-based inference methods described in Section 2.1.1.

Current deep learning architectures have shown capabilities in improving image resolution while facing inherent limitations regarding reliability and authenticity. These approaches introduce specific challenges through potential hallucination effects (where AI models generate plausible but non-existent features) and memorization effects (where models reproduce training patterns rather than generalizing effectively).

Research has established fundamental limitations in the scaling capabilities of super-resolution models. Recent reviews and empirical studies in the field of super-resolution (e.g., Wang et al., 2020; Yang et al., 2019) have established that performance tends to degrade significantly as scaling factors increase. This is due to the growing loss of information in the low-resolution input and the increased risk of generating hallucinated or non-existent details, which undermines the reliability of the enhanced output. For satellite imagery, Lanaras et al. (2018) similarly note that scaling factors above 4× are generally avoided due to these reliability concerns, with the magnitude and characteristics of this degradation varying based on multiple influential factors as discussed in Section 3.2.1.

Specialized architectures have emerged to address constraints specific to satellite imagery applications, incorporating techniques such as physics-guided training to mitigate atmospheric effects and preserve spectral relationships. Current development trajectories focus on model stability enhancement, computational efficiency, and physics-guided approaches.

3 Fundamental Constraints and Technical Boundaries in AI Super-Resolution Technology

The application of AI-SR technology to Sentinel-2 imagery operates within specific technical constraints that shape its utility for CAP monitoring applications. Understanding these performance boundaries and the mitigation strategies available enables paying agencies to make informed implementation decisions while demonstrating compliance with the "equivalent value" standard established in Implementing Regulation (EU) 2022/1173. This section examines the scaling constraints that influence enhancement reliability alongside the validation frameworks and decision protocols that can address these limitations within regulatory monitoring contexts.

3.1 Scaling Limitations

As established in Section 2.2, AI-SR performance exhibits progressive degradation as enhancement factors increase, with the empirical guideline of 4×enhancement. When operating at higher enhancement factors, AI models may generate significantly more hallucination effects - visually convincing and contextually plausible but factually incorrect features - that could compromise monitoring reliability, and potentially affecting the accuracy of eligibility assessment or compliance monitoring.

Nevertheless, this 4× threshold represents a general guideline rather than an absolute limit. The relationship between enhancement factors and output reliability follows a non-linear pattern influenced by several variables:

— **Information Limitation and Exponential Scaling Constraints**: Image resolution enhancement inherently creates quadratic scaling in synthetic content generation. As enhancement factors increase, AI models must generate progressively more synthetic content based on increasingly limited source data. The mathematical relationship is straightforward: each enhancement factor requires quadratic growth in pixel generation. For example, a 4× enhancement requires generating 16 pixels (4×4 grid) from each original pixel. This quadratic scaling relationship, combined with the diminishing information foundation, increasingly challenges output reliability as applied scaling factors increase.

— **Image Content Complexity**: Enhancement performance characteristics vary significantly depending on landscape characteristics and observation conditions:

- Homogeneous agricultural landscapes with regular patterns may support higher enhancement factors (e.g., large cereal fields, uniform vineyard rows, extensive monoculture crops)
- Areas with distinct boundaries and strong contrast features demonstrate better enhancement performance (e.g., field edges adjacent to roads, crop boundaries against water bodies, forest-cropland transitions)
- Heterogeneous regions containing diverse landscape features, fine-scale elements, or complex textures typically exhibit more pronounced deterioration at higher enhancement factors (e.g., mixed farming systems, small-parcel agriculture, agroforestry areas, complex mosaic landscapes), with operational implications for boundary delineation illustrated in Section 5.1 (Figure 3)

- Seasonal variations and atmospheric conditions affect enhancement reliability across temporal sequences (e.g., varying cloud cover, seasonal haze, crop phenology stages, changing illumination angles)

— **Architectural Constraints**: Higher enhancement factors require increasingly complex neural network architectures to generate the additional synthetic content. These deeper, more complex networks are more difficult to train reliably and more prone to generating artifacts, as detailed in Annex 1.1.

— **Training Dataset Composition and Representativeness**: The scope, quality, and diversity of training datasets directly influence model generalisation capabilities at higher enhancement factors. Models trained on datasets with insufficient geographical coverage, seasonal diversity, or spectral variability may exhibit reduced reliability when applied to diverse operational contexts, with performance characteristics varying significantly across different enhancement factors and application domains. These theoretical constraints manifest observably in operational implementations, particularly when enhancement algorithms encounter features that diverge from typical training data distributions. Figure 2 provides empirical evidence of this phenomenon through comparative analysis of photovoltaic infrastructure - a relatively recent addition to agricultural landscapes that may be underrepresented in training datasets compiled from historical imagery.

Figure 2: Comparison of very high-resolution reference imagery from ESRI base map (left) with Sentinel-2 AI-SR enhanced output (right) for agricultural parcels containing photovoltaic installations. The left panel displays the ground reference condition where solar panel arrays are clearly visible as regular geometric patterns with distinct boundaries. The right panel shows the same area as processed through AI-SR enhancement of Sentinel-2 imagery.

Source: © 2025 DigiFarm AS[1] - All Rights Reserved

[1] The example provided by DigiFarm AS serves an educational purpose, representing the operational output of a technology undergoing continuous development and refinement. The authors acknowledge that AI-SR technology is a rapidly evolving field where today's constraints may be addressed through tomorrow's innovations. This example should not be interpreted as an evaluation of a specific commercial product, but rather as an illustrative case that enhances understanding of fundamental AI-SR operational principles.

The comparison between reference imagery and AI-SR output in Figure 2 reveals how statistical inference-based enhancement responds to underrepresented features. The algorithm, lacking sufficient training examples of photovoltaic installations with their distinctive linear arrangements and uniform reflectance patterns, reconstructs these areas using learned representations from more prevalent agricultural features. This manifests as observable smoothing and textural modification - the sharp geometric boundaries and regular panel arrays clearly visible in the reference imagery are replaced with organic patterns more consistent with “traditional” agricultural land cover.

This transformation exemplifies a fundamental principle in AI-SR implementation: enhancement reliability directly correlates with training data representativeness. While the enhanced imagery maintains visual plausibility and general landscape context, specific details essential for accurate land use documentation may undergo alteration in the enhanced output, potentially affecting land use classification and monitoring accuracy.

Contemporary commercial AI-SR solutions actively address this challenge through iterative training data expansion, systematically incorporating emerging agricultural infrastructure patterns as they become more prevalent in the landscape. This ongoing adaptation represents the technology's current maturation phase, where the temporal lag between infrastructure innovation and training data incorporation creates temporary but manageable limitations in reconstruction accuracy.

3.2 Validation Requirements

The synthetic nature of AI-SR outputs necessitates structured validation protocols proportional to the enhancement factor applied. Implementation requires:

— **Systematic Artifact Detection:** Methodologies for identifying potential hallucination effects through both computational metrics and expert analysis, as described in Section 6.3: Artifact Detection and Validation Methodology and further detailed in Annex 4.1: Metric-Based Detection Approaches for Certification Requirements.

— **Multi-Source Verification:** Validation of enhanced imagery through complementary reference data sources that may include native Very High Resolution (VHR) satellite imagery (0.3-0.5m), orthorectified aerial photography, ground-based observations, and where available, LiDAR or hyperspectral data. This verification approach enables identification of potential divergence between AI-generated content and actual ground conditions. The technical architecture for reference data integration is documented in Annex 4.5: Reference Data Integration.

— **Decision Framework for Uncertainty Management:** An analytical approach for addressing cases where enhanced imagery exhibits potential artifacts or ambiguities. This framework may incorporate tiered assessment protocols based on the criticality of the monitoring task and the degree of uncertainty in the enhanced outputs, as detailed in Annex 4.3: Metric Aggregation and Anomaly Prioritization for Certification Workflows.

When validation reveals ambiguities affecting eligibility assessment or compliance monitoring, native VHR imagery or field observations serve as authoritative references.

The validation methodologies outlined above establish a technical framework for maintaining evidentiary integrity within the CAP monitoring framework. These methodologies form the technical foundation for the quality assurance protocols detailed in Section 6: Evidential Value and Certification Framework.

4 Implementation Considerations: A Structured Evaluation Approach

Super-resolution technology constitutes a rapidly evolving domain with applications extending far beyond agricultural contexts. The field benefits from substantial research contributions across multiple disciplines including computer vision, machine learning, and remote sensing. This breadth of development establishes a comprehensive technical foundation for evaluating potential relevance to CAP monitoring while introducing specific considerations for quality assurance integration and operational reliability that paying agencies should assess.

The technical landscape has evolved from primarily academic research to operational applications. This evolution has been facilitated by significant advances in deep learning architectures and increasing availability of high-performance computing resources, both factors influencing evaluation and potential implementation decisions for paying agencies considering AI-SR technology.

4.1 Strategic Evaluation: Decision Framework and Resource Considerations

Integration of AI-based super-resolution technology within CAP monitoring systems necessitates systematic evaluation of both strategic implications and implementation pathways. Such consideration represents a significant organisational decision requiring comprehensive assessment of institutional capabilities and resource requirements. Developing specialized expertise for managing AI-enhanced imagery and associated quality control protocols remains essential to demonstrate the 'equivalent value' established in Implementing Regulation (EU) 2022/1173.

When evaluating AI-SR technology, paying agencies should consider the fundamental relationship between enhancement factors and output reliability established in Section 3.1. Higher enhancement factors increase synthetic content generation, potentially compromising evidential value and quality assurance standards. This relationship necessitates structured risk management frameworks in any evaluation process.

In evaluating AI-SR technology, paying agencies can assess two distinct implementation approaches, each presenting different trade-offs regarding technical control, resource requirements, and implementation timelines:

— **Open-source frameworks** provide greater technical control and customisation potential but require substantial internal expertise for development, testing, and maintenance. These solutions offer maximum flexibility but demand significant technical resources throughout the implementation lifecycle. Implementation involves considerable investment in specialized staff competencies across deep learning, remote sensing, and large-scale data processing. For detailed technical specifications and resource requirements of open-source solutions, please refer to Annex 3.2: Open-Source Solutions: Capabilities and Resource Requirements.

— **Commercial solutions** offer more immediate implementation potential with reduced technical overhead, though potentially limiting customisation options and future adaptability. These solutions generally provide streamlined integration with existing monitoring systems but may introduce vendor dependencies. Commercial solutions typically involve lower initial implementation complexity but may entail higher long-term operational costs through service agreements. For detailed analysis of service models and integration approaches for commercial solutions, please refer to Annex 3.3: Commercial Solutions: Service Models and Integration Approaches.

The assessment of these potential implementation approaches would benefit from systematic analysis of:

— **Technical capabilities**: Current institutional expertise versus required competencies

— **Resource availability**: Budget constraints, infrastructure capacity, and staff resources

— **Operational requirements**: Monitoring priorities, implementation timelines, and performance expectations

— **Risk management framework**: Processes for managing uncertainty in enhanced imagery

— **Long-term sustainability**: Continued maintenance, periodic revalidation, and technical evolution

AI-SR integration requires a hierarchical decision framework providing clear protocols for resolving ambiguities in enhanced imagery. Such a framework needs to explicitly designate original high-resolution imagery or field observations as authoritative references when enhanced imagery exhibits uncertainty in areas affecting eligibility or compliance assessment.

Resource planning should consider the complete lifecycle of potential implementation, including:

— **Infrastructure requirements**: Computing resources and storage systems

— **Personnel development**: Training programs, expertise acquisition, and operational staff

— **Quality control systems**: Validation frameworks, uncertainty management, and audit protocols

— **Documentation systems**: Metadata management, processing records, and traceability mechanisms

For detailed implementation guidance, please refer to Annex 3.1.

4.2 Regulatory Compliance and Certification Considerations

AI-SR technology adoption necessitates integration within quality assurance frameworks. Building upon the paradigm shift considerations established in Section 2.1.2, this report recommends implementing certification frameworks that serve dual purposes: distinguishing between native and AI-enhanced imagery, and ensuring that any synthetic content undergoes systematic quality control to demonstrate compliance with the "equivalent value" standard established in Implementing Regulation (EU) 2022/1173. For AI-enhanced imagery, achieving this regulatory standard involves systematic validation protocols that verify enhanced outputs maintain comparable reliability to established monitoring methodologies, thereby supporting the evidential integrity essential for regulatory compliance.

Implementation involves certification methodologies that support Member States' quality assurance systems by establishing:

- **Authentication** protocols for distinguishing native from AI-enhanced imagery
- **Validation** methodologies for systematic detection of potential artifacts
- **Documentation** standards maintaining complete processing records
- **Chain of evidence** procedures ensuring traceability for regulatory purposes

The certification and validation frameworks establish the evidential value of enhanced imagery within quality assurance systems, ensuring reliable utility for compliance monitoring and eligibility determination. The certification framework outlined in Section 6 provides an analytical framework for establishing the evidential value of enhanced imagery.

Technical implementation details are provided in Annex 4.

4.3 Emerging Validation and Certification Research

Technological evaluation must consider current capabilities and limitations, while acknowledging ongoing research suggesting potential future advancements in certification and validation methodologies that may eventually address limitations discussed in Section 6 on evidential value and certification. These developments remain in research stages with potential operational readiness timelines extending beyond 2026-2027. Near-term assessment should therefore focus on the certification frameworks and validation methodologies detailed in Section 6.3 on artifact detection and validation methodology.

Further details are provided in Annex 5.

4.4 Framework Evaluation Criteria

Integration of AI-SR within CAP monitoring systems requires a systematic approach to evaluating potential implementation frameworks, balancing technical capabilities, resource constraints, and operational requirements. This evaluation examines:

- **Enhancement factor considerations**: Implementation decisions should account for the scaling constraints detailed in Section 3.2, with particular scrutiny for applications requiring enhancement factors beyond practical reliability thresholds
- **Infrastructure integration assessment**: Evaluation would need to examine compatibility with data management systems, processing workflows, and metadata architectures.
- **Validation and certification framework compatibility**: The methodologies outlined in Section 6.3 on artifact detection and validation methodology establish a foundation for identifying potentially erroneous synthetic features.
- **System adaptability and future readiness**: This includes consideration of modular architectures that could accommodate improved validation techniques and enhanced processing capabilities as they become available.

For detailed implementation frameworks, please refer to Annex 3.

5 Operational Impacts of AI Super-Resolution Technology on CAP Monitoring

This section examines the practical implications of AI-SR's technical characteristics for key monitoring activities, building upon the performance boundaries established in Section 3.2 and the implementation frameworks outlined in Section 4, with particular attention to three critical areas of CAP monitoring.

5.1 Production and Update of Parcel Borders

Maintaining accurate and up-to-date parcel boundaries through the Land Parcel Identification System (LPIS) constitutes a foundational element of the CAP implementation. Current approved approaches rely on high-resolution imagery (<0.5 m) as authoritative input for the Geo-Spatial Application System (GSA).

When considering AI-SR technology for parcel delineation, technical parameters require evaluation: Sentinel-2's baseline 10-meter resolution would require a scaling factor of approximately 20× to approach the 0.5-meter resolution required for accurate boundary delineation. This substantially exceeds the reliability constraints discussed in Section 3.2.1, requiring generation of 400 synthetic pixels (20×20 grid) from each original pixel, impacting output evidential value and quality assurance standards.

Precise delineation serves two distinct but equally critical functions:

— **Parcel boundary identification**: Defining the outer limits of agricultural parcels eligible for CAP payments

— **Ineligible feature detection**: Identifying features within parcels that must be excluded from payment area calculations (e.g., water bodies)

In that context, the synthetic nature of AI-enhanced imagery introduces significant evidentiary considerations as the enhancement process can generate hallucinated features that create uncertainties about:

— **Commission errors**: The presence of ineligible features in the enhanced image that do not actually exist on the ground

— **Omission errors**: The unintentional replacement of ineligible features with eligible ones

— **Boundary distortion**: Inaccurate parcel boundary representation affecting area measurements

These uncertainties may impact the accuracy of eligible area calculations and, consequently, the areas reported for the interventions under CAP regulations.

Boundary distortion becomes particularly evident when AI-SR processes interfaces between spectrally similar land cover types. Figure 3 provides empirical evidence of this challenge through comparative analysis of forest-grassland boundaries, demonstrating how enhancement processing may affect the precise delineation critical for parcel identification.

Figure 3: Comparison of very high-resolution reference imagery from ESRI base map (left) with Sentinel-2 AI-SR enhanced output (right) depicting forest and grassland parcels. The left panel displays clear boundary delineation between forest and grassland at sub-meter resolution. The right panel presents the same area after AI-SR enhancement of Sentinel-2 imagery, illustrating characteristic boundary interpretation when processing spectrally similar land covers.

Figure 3 illustrates how AI-SR algorithms interpret transitional zones between spectrally similar land covers through statistical inference. In such cases, the enhancement process may generate graduated boundaries that differ from the discrete ecological transitions visible in reference imagery. For parcel boundary identification - where precise delineation determines eligible areas for CAP payments - these processing characteristics, when they occur, introduce operational considerations that paying agencies can address through appropriate validation protocols.

To mitigate these challenges and enhance the reliability of AI-SR for parcel delineation, paying agencies may consider AI-SR technology for enhancing imagery from sources with resolution closer to the required standard. For example, moderate-resolution satellite imagery with a native resolution of 1 meter could be enhanced with scaling factors of 2× to approach the required resolution while operating within the reliability parameters discussed in Section 3.2.1. This application would require thorough validation to demonstrate compliance with quality assurance requirements in operational contexts.

Additionally, strategic acquisition timing - selecting periods when land cover types exhibit maximum spectral contrast, such as autumn when deciduous forests and grasslands are most distinct - may improve boundary definition in both source imagery and subsequent AI-SR processing.

5.2 Identification and Monitoring of Landscape Features

Small Landscape Features (SLFs) play a crucial role in biodiversity conservation and ecological balance within CAP's environmental objectives. The fundamental challenge in landscape feature monitoring stems from their often-limited spatial extent, complex morphology, and variable spectral signatures - characteristics that create specific detection challenges.

This is particularly significant for features that approach or fall below the native resolution of Sentinel-2 imagery, where enhancement algorithms must generate synthetic content based on

[2] DigiFarm AS examples shown for educational purposes; see Figure 2, footnote 1 for full disclaimer.

limited information in the original data. These technical constraints manifest in landscape monitoring through several distinct mechanisms:

— **Substitution Effects**: AI-SR may transform actual landscape elements into different feature types based on statistical patterns rather than actual characteristics (as illustrated with photovoltaic infrastructure in Section 3.1, Figure 2), creating challenges for habitat classification.

— **Commission Errors**: AI-SR may generate contextually plausible but non-existent landscape features that appear consistent with surrounding ecological patterns, creating artificial habitat continuity.

— **Omission Errors**: Enhancement processes may fail to preserve actual small landscape features, particularly those with limited contrast against surrounding vegetation, artificially simplifying landscape complexity. These omission effects are particularly evident for linear landscape features that fall near the resolution threshold of the source imagery. Figure 4 illustrates this issue through the inadequately reconstructed tree row in AI-SR enhanced imagery, showing how structural definition essential for feature identification has been significantly reduced.

Figure 4: Comparison of very high-resolution reference imagery from ESRI base map (left) with Sentinel-2 AI-SR enhanced output (right) showing agricultural parcels with a linear tree feature. The left panel clearly displays a mature tree row forming a field boundary with distinct canopy structure. The right panel shows the same area after Sentinel-2 AI-SR enhancement where the tree row's linear structure and defining characteristics have been impaired.

Source: © 2025 DigiFarm AS3 - All Rights Reserved.

The reconstruction challenges illustrated in Figure 4 exemplify a significant consideration for SLF monitoring. While AI-SR processing attempts to preserve linear features, the resulting output may lack the spatial precision necessary for reliable identification. Features such as tree rows, hedgerows, and buffer strips may become ambiguous in their ecological function and regulatory

[3] DigiFarm AS examples shown for educational purposes; see Figure 2, footnote 1 for full disclaimer.

status, presenting particular challenges for automated detection systems where partial visibility impedes confident classification and compliance assessment.

The operational implications extend beyond technical assessment to encompass evidential value and regulatory considerations. Inaccurate feature detection may affect compliance determination with environmental conditionality criteria and eligible area calculations. Conversely, incomplete detection of existing landscape features could impact the protection of ecologically significant elements, potentially affecting CAP's environmental objectives.

For paying agencies considering AI-SR implementation, these challenges necessitate robust safeguards, with particular emphasis on the validation protocols established in Section 6.3. The resource requirements for appropriate safeguards represent an important consideration when evaluating the overall operational benefits of AI-SR technologies.

5.3 Agricultural Practices Monitoring

Sentinel-2's primary monitoring strength derives from its high temporal frequency (detailed in Annex 2), enabling robust time-series analysis that compensates for moderate spatial resolution in specific applications. The value proposition of AI-SR should therefore be evaluated within this context of existing temporal capability.

For large-scale agricultural practices monitoring - including crop diversification, winter soil cover, conventional tillage operations, catch crops, nitrogen-fixing crops, and fallow land management - temporal analysis of native resolution Sentinel-2 imagery, potentially combined with Sentinel-1 data, provides sufficient information for monitoring purposes. These practices typically exhibit distinctive spectral-temporal signatures effectively captured at current resolution. For parcels of adequate size, higher spatial resolution imagery may be less critical, potentially reducing the need for spatial resolution enhancement.

For agricultural parcels too small to be effectively monitored with Sentinel-2 imagery, AI-SR offers limited advantage due to spectral mixing. The mixed spectral signatures from adjacent land covers within individual pixels cannot be reliably unmixed through AI-SR techniques, as the algorithm lacks the information necessary to accurately disaggregate these composite signals.

Within the CAP monitoring framework, fine-scale agricultural elements present distinct analytical challenges. For practices requiring high spatial resolution monitoring (e.g., buffer strips, landscape features, and riparian buffer zones), enhanced spatial detail would theoretically offer monitoring benefits. However, these applications would necessitate enhancement factors that substantially exceed the reliability boundaries discussed in Section 3.1, making native VHR imagery (0.3-0.5m) or field observations more practical alternatives given the extensive validation requirements needed to establish the evidential value of such highly enhanced imagery.

The comprehensive validation requirements detailed in Section 4.2 represent an important factor in evaluating the overall value proposition for these specific monitoring applications.

5.4 AI-Based Super-Resolution Integration within CAP Monitoring Systems: Technical and Operational Framework

The integration of AI-SR technology into existing CAP monitoring systems represents a strategic decision for paying agencies that involves systematic coordination across regulatory, technical, and operational dimensions to safeguard monitoring reliability and the integrity of the CAP framework.

Meeting regulatory requirements is the primary consideration for AI-SR implementation, as it directly affects whether enhanced imagery can serve as acceptable evidence in CAP monitoring. Integration involves certification protocols and a hierarchical evidence framework, designating native imagery (unenhanced) or field observations as authoritative references while defining specific validation requirements for enhanced imagery to attain evidentiary status. Implementation should incorporate the certification framework detailed in Section 6 to ensure enhanced imagery maintains appropriate evidential value for CAP monitoring.

At the technical level, integration with LPIS and GSA systems necessitates architectural considerations that account for the relationship between enhancement factors and output reliability established in Section 3.2. Technical implementation should incorporate enhancement factor constraints ensuring reliability remains demonstrably high, while maintaining comprehensive documentation of enhancement parameters, model characteristics, and processing decisions as outlined in Section 6.5 on metadata architecture and traceability protocols.

The operational dimension focuses on process adaptation that acknowledges the fundamental distinction between traditional satellite imagery (direct optical capture) and AI-enhanced imagery (partially synthetic representation).

Effective implementation requires phased deployment, beginning with low-risk applications and expanding to comprehensive monitoring functions as the system demonstrates reliability. This integration process demands a holistic approach that prioritizes regulatory compliance while optimizing technical and operational elements to preserve the fundamental integrity of CAP monitoring processes.

Ultimately, paying agencies retain autonomy in determining appropriate implementation pathways, tailored to their specific operational contexts, and risk management frameworks, while always prioritising the robust maintenance of regulatory standards.

6 Evidential Value and Certification Framework

Implementing Regulation (EU) 2022/1173 permits Member States to utilize data of *"at least equivalent value"* for Area Monitoring System applications (Articles 10(3) and 11), with data quality adequacy determined through Member States' established quality assurance processes.

This section proposes a certification framework that supports Member States in establishing the evidential value of AI-SR imagery within their quality assurance systems. The framework provides structured technical guidance for implementation.

6.1 The Changing Nature of Image Evidence

Building on the paradigm shift established in Section 2.1.2, the practical implications of this transition from direct capture to synthetic content within regulatory contexts manifest through specific mechanisms detailed in the technical limitations established in Section 3.2 on performance boundaries and scaling constraints:

AI enhancement introduces inherent uncertainties through potential hallucination and memorization effects, where the enhancement process may generate features that appear contextually plausible but do not correspond to actual ground conditions

Modern AI-enhanced imagery has reached a level of sophistication where distinguishing between native and enhanced imagery through visual inspection alone has become increasingly challenging.

This transformation highlights the importance of three important considerations: a robust certification framework to distinguish native from AI-enhanced imagery (Section 6.2 on certification requirements), comprehensive technical validation protocols (Section 6.3 on artifact detection and validation methodology), and clear quality requirements for enhanced imagery's evidential status (Section 6.4 on Quality Assurance Framework). These elements represent important considerations for maintaining reliable image-based evidence in CAP monitoring, where accuracy directly influences regulatory and financial outcomes.

6.2 Certification: Establishing Evidence Reliability

With the ongoing advancement of AI-based image processing technologies, certification offers pathways for establishing the evidential value of imagery used for compliance verification. A certification framework would establish either that an image has maintained its original characteristics without any AI enhancement, or that any AI enhancement applied has been systematically validated to ensure sufficient reliability relative to actual ground conditions.

This is particularly important for CAP implementation, where imagery serves as primary evidence for regulatory compliance and financial decisions; Certification approaches would strengthen, or re-establishes, trust in the imagery used, enabling its continued use in regulatory frameworks while fostering confidence in robust CAP implementation.

6.3 Artifact Detection and Validation Methodology for AI-Enhanced Imagery

Identifying artifacts in AI-enhanced imagery is technically challenging due to the inherent nature of AI super-resolution technology. As discussed in Section 2.1.1, contemporary AI-SR architectures are specifically designed to generate photorealistic details that appear convincing to human perception but may not always represent actual ground conditions. This creates a complex validation problem: the visually plausible nature of AI-generated artifacts makes them exceedingly difficult to identify through conventional visual inspection, particularly when operating at scale across vast and heterogeneous agricultural landscapes. This validation challenge is not merely theoretical. As demonstrated in Section 3.1's photovoltaic infrastructure analysis, contemporary AI-SR systems can generate compelling visual representations that misrepresent actual ground features.

This technological challenge directly drives the need for the certification requirements established in Section 6.2 as the reliability of enhanced imagery depends fundamentally on distinguishing between authentic features and AI-generated artifacts. Establishing the evidential value of enhanced imagery through systematic validation protocols enables demonstration of the 'equivalent value' required under Implementing Regulation (EU) 2022/1173.

A robust methodological framework addresses these challenges through the integration of quantitative metrics with systematic expert analysis across three interconnected components:

- **Metric-Based Detection**: Specialized metrics detect anomalies that might escape human perception by evaluating structural patterns, perceptual characteristics, and statistical properties of enhanced imagery. These metrics provide quantitative assessment that can direct specialist attention toward regions requiring detailed evaluation.
- **Prioritization**: A systematic aggregation framework transforms multiple metric outputs into structured information that highlights potential artifacts. This process establishes a multi-tiered review framework (standard monitoring, enhanced verification, and critical investigation) based on statistical thresholds, enabling efficient allocation of expert analysis resources.
- **Expert-Driven Visual Analytics**: Interactive visualization systems enable experts to conduct detailed examination of prioritized regions through multi-source reference data integration, synchronised temporal analysis, and comprehensive documentation capabilities.

These components work together to establish the evidentiary value of enhanced imagery, ensuring that CAP monitoring systems maintain their reliability while potentially benefiting from resolution enhancement capabilities. Through structured validation procedures, the framework creates a systematic path for enhanced imagery to establish regulatory credibility - translating enhanced visual detail into reliable evidence for compliance verification and eligibility determinations.

This approach enables paying agencies to maintaining regulatory compliance. The system fundamentally recognises the irreplaceable value of human expertise in artifact validation while providing Visual Analytics tool to augment and focus expert analysis. The framework incorporates multi-source reference data, including satellite imagery, orthophotos, LiDAR data, and ground-level photography to enable comprehensive verification of enhanced imagery.

For comprehensive technical specifications of the artifact detection and validation methodology, including detailed descriptions of metric-based approaches, aggregation frameworks, visual analytics systems, and reference data integration, please refer to Annex 4: Technical Specifications.

6.4 Quality Assurance Framework: Evidential Value and Compliance Standards

The quality assurance framework should now account for this new reality in image-based evidence. It provides specific recommendations for CAP implementation, defining guidelines for image quality and accuracy regardless of the processing method. The framework incorporates multiple verification layers, including automated systems and expert reviews.

Expert testimony takes on increased importance, with technical experts needed to communicate not only about image enhancement processes but also about certification procedures and validation results. The documentation creates a complete audit trail that includes:

- Systematic identification of any AI enhancement applied
- For in-house implementations: Model specifications and performance metrics; For commercial services: Provider identification, algorithm version, service specifications, and available performance metrics
- For in-house implementations: Training data characteristics and detailed validation results; For commercial services: Available quality assessments, validation results, and/or provider certifications
- Processing chain documentation
- Quality control measures and results

6.5 Metadata Architecture: Documentation and Traceability Protocols

Comprehensive metadata management constitutes a critical component of a certification framework. Its importance stems from two fundamental requirements: maintaining complete traceability of enhancement processes for quality assurance compliance and enabling systematic diagnostic procedures when enhanced outputs exhibit unexpected behavioral patterns or deviate from expected quality metrics.

The metadata framework documents two essential domains. First, enhancement process documentation record model specifications and processing parameters, including specific settings that address the fundamental limitations discussed in Section 3.2 on performance boundaries and scaling constraints. Second, quality control documentation maintains detailed records of validation processes, reference data sources, and decision criteria applied.

This metadata management approach enables rapid identification of potential issues and maintains audit trails for compliance requirements. It provides essential documentation for expert testimony within quality assurance frameworks, as discussed in Section 6.4.

7 Conclusions

While Sentinel-2 imagery offers valuable temporal coverage and spectral capabilities for agricultural monitoring, its 10-meter spatial resolution presents a constraint for specific applications. This technical gap has driven the evaluation of AI-based Super-Resolution as a potential enhancement, necessitating a systematic analysis of its capabilities and implementation within CAP regulatory frameworks.

AI-SR enables spatial resolution enhancement while fundamentally transforming the evidential nature of satellite imagery. AI-SR implements statistical inference to generate realistic details. This process transforms satellite imagery from direct optical documentation of physical reality to enhanced representation that incorporates synthetically generated content. This represents a paradigm shift where satellite imagery transitions from being inherently trustworthy evidence of ground conditions to partially synthetic representation that requires validation to establish its evidential reliability.

Evidential value decreases as enhancement factors increase because higher scaling factors require generating progressively more synthetic content from increasingly limited original information. Reliability is influenced by multiple interdependent factors: enhancement magnitude, landscape complexity, training dataset representativeness, atmospheric conditions, and temporal variations. A 4× threshold serves as general technical guidance where these various factors typically converge to maintain acceptable performance boundaries.

Establishing the evidential value of AI-enhanced imagery becomes essential for regulatory compliance, particularly to address the requirement for "*at least equivalent value*" specified in Implementing Regulation (EU) 2022/1173. The certification framework proposed provides structured methodologies for distinguishing between native satellite imagery and AI-enhanced outputs, while systematically validating enhanced imagery evidential value. This framework integrates artifact detection protocols, identifying potential hallucination and memorisation effects, along with multi-source verification procedures using reference datasets, and structured uncertainty management protocols.

Implementation of AI-SR technology within CAP monitoring requires comprehensive metadata architecture and traceability protocols to maintain transparency and accountability throughout the enhancement process. Quality assurance systems should systematically document enhancement parameters, model specifications, processing decisions, and validation outcomes. The proposed framework supports both operational quality control and regulatory compliance, ensuring that enhanced imagery maintains appropriate evidential standards when used for compliance verification or eligibility determinations.

The report provides paying agencies with an evaluation framework for systematically assessing AI-SR technology within their specific operational contexts. The examination encompasses technical foundations, performance boundaries, implementation pathways, operational impacts, certification requirements, and deployment strategies, enabling informed decision-making based on technical capabilities, resource requirements, and regulatory compliance considerations.

Paying agencies are empowered with complete implementation autonomy, with this analytical framework serving as technical guidance, recognising that optimal implementation approaches correspond to institutional capabilities, monitoring requirements, and operational constraints.

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List of abbreviations and definitions

Abbreviations	Definitions
AI	Artificial Intelligence: Computational systems that simulate human cognitive functions using algorithms and neural networks.
CAP	Common Agricultural Policy: the EU's agricultural framework implementing market interventions, direct payments, and rural development programs while establishing regulatory standards for agricultural practices.
CNN	Convolutional Neural Network: deep learning architecture utilizing convolution operations and hierarchical feature extraction for processing grid-structured data, optimized for computer vision tasks.
ESSIM	Enhanced Structural Similarity Index: Advanced image comparison metric that evaluates structural elements, textures, and contrast patterns to assess image quality and detect artifacts in enhanced imagery.
FPS	Feature Persistence Score: Validation metric that quantifies the consistency of landscape features across different data sources and temporal conditions to identify potential AI-generated artifacts.
GAN	Generative Adversarial Network: Machine learning framework where generator and discriminator networks compete to produce synthetic images indistinguishable from authentic samples through adversarial training.

Abbreviations	Definitions
GLCM	Gray Level Co-occurrence Matrix: Statistical method for examining texture by considering the spatial relationship of pixels, used in remote sensing applications for analyzing pattern distributions and detecting anomalies in enhanced imagery.
GSA	Geo-Spatial Aid Application System: Digital platform that allows farmers to graphically declare and visually indicate agricultural areas for which they apply for CAP aid, integrated with LPIS for verification and payment processing.
IACS	Integrated Administration and Control System: EU-wide digital system managing and controlling CAP payments to farmers through interconnected databases including LPIS and GSA, ensuring standardized payment administration across Member States.
LFAS	Local Feature Alignment Score: Metric that evaluates the spatial coherence of landscape features with their surroundings, focusing on the natural alignment of elements such as hedgerows, streams, and field boundaries within the broader landscape context.
LPIS	Land Parcel Identification System: EU's geographical information system designed as the main instrument for implementing CAP direct payments, identifying and quantifying agricultural land eligible for payments through spatial parcel registration and update.
PHD	Perceptual Hash Distance: Validation metric that creates unique pattern fingerprints to detect unnaturally repeated or copied features in AI-enhanced imagery.

Abbreviations	Definitions
RSR-TransUNet	Hybrid neural architecture combining transformer mechanisms with U-Net structure for multi-spectral satellite imagery super-resolution, preserving spectral-spatial information.
SGGAN	Satellite-Guided GAN: GAN variant incorporating physics-guided training for satellite imagery enhancement, ensuring physically consistent results while preserving spectral characteristics
SLFs	Small Landscape Features: Landscape elements including hedgerows, ditches, and isolated trees that enhance biodiversity and ecosystem services.
SR	Super-Resolution: Computational techniques for enhancing image resolution through advanced interpolation and reconstruction methods, utilizing deep learning or physics-based approaches.
SRCNN	Super-Resolution CNN: Pioneering CNN architecture for super-resolution implementing end-to-end mapping between low and high-resolution images through convolutional layers.
SRGAN	Super-Resolution GAN: GAN-based architecture optimized for photo-realistic image super-resolution, utilizing perceptual loss functions and adversarial training.
TCI	Texture Coherence Index: Statistical metric using GLCM analysis to assess texture pattern consistency and identify artificial uniformity in landscape imagery.
TLIR	Transformer Lightweight Image Restoration: Efficient transformer architecture for remote sensing image restoration, employing lightweight self-attention mechanisms for computational efficiency.

Abbreviations	Definitions
VDSR	Very Deep Super Resolution: Deep learning architecture utilizing residual connections and very deep networks (20+ layers) for enhanced detail reconstruction in super-resolution tasks.
VHR	Very High-Resolution: Satellite imagery with spatial resolution finer than 1 meter per pixel, enabling detailed analysis of small-scale ground features.

List of figures

Annexes

This section provides a systematic guide to the technical annexes supporting this report. The annexes follow a progressive structure from foundational concepts to implementation frameworks, certification methodologies, and future research directions. Each annex addresses specific dimensions of AI-SR technology within CAP monitoring contexts, documenting technical specifications, implementation parameters, and analytical frameworks relevant to regulatory considerations.

Annexes 1 & 2 document the technical foundations of AI super-resolution and Sentinel-2 platform specifications.

Annex 3 examines the implementation frameworks, including strategic decision frameworks, open-source solutions, and commercial options.

Annex 4 details certification and validation methodologies for establishing the evidentiary value of enhanced imagery.

Annex 5 explores emerging validation technologies with implementation timelines beyond 2026-2027.

These annexes are systematically referenced throughout the main text to provide access to detailed technical information as needed, while maintaining the strategic focus of primary analysis.

Annex 1. Technical Foundations: Deep Learning Architectures for Super-Resolution

This annex provides technical details on the evolution of deep learning frameworks used in super-resolution technology. It documents specific neural network architectures, their operational principles, and characteristic constraints that shape implementation possibilities. This technical foundation is essential for understanding the capabilities and limitations of AI-SR within agricultural monitoring applications.

Evolution of Deep Learning Architectures for Super-Resolution

The progression of deep learning approaches represents a fundamental transformation in super-resolution technology, introducing new capabilities while establishing inherent limitations that directly impact regulatory applications. While these architectures demonstrate significant advancement in resolution enhancement, they simultaneously introduce specific constraints through hallucination effects (generation of plausible but non-existent features) and memorization effects (reproduction of training patterns rather than effective generalization).

The foundation of contemporary AI-SR technology emerged with Dong et al.'s (2014) introduction of the Super-Resolution Convolutional Neural Network (SRCNN). This pioneering architecture demonstrated that convolutional neural networks could learn complex mapping functions between low-resolution and high-resolution images, effectively modeling relationships between corresponding resolution pairs. Kim et al. (2016) advanced this approach with the Very Deep Super Resolution (VDSR) network, implementing residual learning to focus specifically on high-frequency details like textures and edges while maintaining structural fidelity.

A significant paradigm shift occurred with Ledig et al.'s (2017) introduction of SRGAN (Super-Resolution Generative Adversarial Network), which applied Goodfellow et al.'s (2014) GAN framework to super-resolution tasks. SRGAN's dual-network architecture fundamentally transformed the enhancement approach:

— The **generator network** transforms low-resolution inputs into higher-resolution outputs through convolutional layers and up-sampling operations, progressively building detail from broad structural elements to finer textures.

— The **discriminator network** evaluates enhanced images by comparing them against authentic high-resolution samples, effectively learning to distinguish between real and synthetically generated imagery.

— The **competitive interaction** between these networks creates a self-improving system where the generator continuously refines its enhancement capabilities to overcome the discriminator's increasingly sophisticated detection abilities. The training process theoretically concludes when the system reaches Nash equilibrium - the point where the generator produces images so convincingly realistic that the discriminator can no longer reliably distinguish them from authentic high-resolution images.

This adversarial training process drives the generator to produce increasingly realistic high-resolution images. SRGAN also introduced perceptual loss functions that evaluate differences in high-level feature representations rather than pixel-by-pixel comparisons. This approach aligns more closely with human visual perception and produces more visually convincing outputs, though not necessarily more accurate representations of ground conditions - a distinction with significant implications for regulatory applications.

Performance Boundaries and Fundamental Limitations

Research has established scaling limitations in super-resolution models that directly impact agricultural monitoring applications. Multiple studies, including Li et al. (2022) and Lanaras et al. (2018) demonstrated that enhancement performance deteriorates significantly as scaling factors increase beyond 4×, creating a technical constraint for implementation within CAP monitoring frameworks. This performance boundary stems from two fundamental limitations:

- **Information Limitation**: Low-resolution input images contain finite information. As enhancement factors increase, the system must generate progressively more synthetic content based on increasingly limited source data, directly affecting output reliability.
- **Architectural Constraints**: Higher enhancement factors require deeper neural network architectures, which introduce training instability and increased potential for hallucination effects, where the model generates contextually plausible but factually incorrect features.

These limitations establish a non-linear relationship between enhancement factors and output reliability, with significantly diminished reliability as enhancement factors increase. This relationship forms a critical technical parameter for regulatory applications where accuracy directly affects compliance assessment and payment determination.

Specialized Architectures for Earth Observation Applications

Recent advances have produced architectures specifically optimized for satellite imagery applications, addressing domain-specific challenges not present in general computer vision tasks. These specialized frameworks implement technical approaches directly relevant to agricultural monitoring applications:

- Wang et al. (2022) Satellite-Guided GAN incorporated physics-guided training to mitigate atmospheric scattering effects, demonstrating improved performance in variable atmospheric conditions.
- Liu et al. (2023) RSR-TransUNet architecture simultaneously preserves both spectral relationships across different wavelength bands and spatial details within each band, maintaining the spectral integrity essential for agricultural classification.
- Zhang et al. (2023) TLIR (Transformer Lightweight Image Restoration) framework implements efficient self-attention mechanisms for computational optimisation in resource-constrained environments.

Current development trajectories focus on three critical technical domains:

- **Model Stability Enhancement**: Addressing the inherent limitations of deeper neural architectures, particularly regarding training stability and artifact reduction.
- **Computational Efficiency**: Developing streamlined architectures and high-performance memory management strategies for practical deployment within infrastructure constraints.
- **Physics-Guided Approaches**: Integrating atmospheric scattering models and seasonal illumination compensation to improve performance across variable environmental conditions.

These specialized architectures provide technical capabilities for enhancing satellite imagery while establishing fundamental constraints on performance reliability. The interplay between enhancement capabilities and reliability limitations directly impacts the evidentiary value of enhanced imagery within quality assurance systems as discussed in Section 6.1 on the changing nature of image evidence.

Annex 2. Sentinel-2 Platform: Specifications and Operational Parameters

This annex provides comprehensive technical specifications of the Sentinel-2 satellite platform and its Multi-Spectral Instrument (MSI). It documents the platform's spatial, spectral, and temporal resolution characteristics that fundamentally shape its capabilities and limitations for agricultural monitoring applications. This detailed examination establishes the technical context necessary for evaluating enhancement requirements and implementation possibilities within CAP monitoring frameworks.

Sentinel-2 constitutes a foundational component of contemporary agricultural monitoring infrastructure, distinguished by its technical architecture and operational framework. Operating within the European Union's Copernicus program, the platform implements an open data policy that fundamentally transforms access dynamics to Earth observation data. This accessibility, combined with systematic acquisition strategies, ensures comprehensive coverage of agricultural areas throughout growing seasons. The constellation achieves a baseline revisit time of 5 days (at the equator) via the combined operation of Sentinel-2A (launched June 2015), Sentinel-2B (March 2017), and the recently launched Sentinel-2C (September 2024). Notably, the revisit time significantly improves over European territories, achieving more frequent revisit of 2-3 days due to orbital convergence at higher latitudes and the resultant overlap between adjacent orbital tracks. This enhanced temporal performance, achieved through coordinated orbital configurations, proves particularly effective for agricultural monitoring across European latitudes.

The Multi-Spectral Instrument (MSI) onboard Sentinel-2 captures electromagnetic radiation through thirteen spectral bands optimized for Earth Observation applications. The instrument's spectral architecture is organised as follows: four bands at 10-meter spatial resolution (covering wavelengths from 443 nm to 945 nm), six bands at 20-meter resolution (spanning from 703 nm to 2190 nm), and three bands at 60-meter resolution (ranging from 443 nm to 1375 nm) specifically designed for atmospheric correction.

The MSI's technical specifications enable comprehensive agricultural monitoring through precisely calibrated spectral responses. The system enables efficient crop classification through the combination of multi-temporal and multi-spectral analysis, facilitates agricultural practice monitoring through temporal analysis, and supports environmental compliance monitoring.

However, the platform's spatial resolution characteristics impose specific operational constraints. While the 10-meter resolution in key bands constitutes a significant advancement in freely available High Resolution Earth Observation data, it proves insufficient for specific CAP monitoring requirements. The resolution limitation affects the monitoring of agricultural parcels below 0.5 hectares and prevents an accurate detection of small landscape features.

These constraints impact CAP implementation capabilities, limiting the accurate detection and measurement of small landscape features and parcel boundaries that require sub-meter observation capabilities, as detailed in Section 4.2 on emerging validation and certification approaches. Consequently, paying agencies need commercial very high-resolution (VHR) satellite imagery (<0.5m resolution) to fulfill regulatory compliance requirements where Sentinel-2's 10-meter resolution proves inadequate.

The interplay between Sentinel-2's specifications and CAP monitoring requirements presents a complex technical landscape. While the platform's spectral bands and temporal resolution enable systematic agricultural monitoring, its spatial resolution characteristics necessitate consideration of enhancement methodologies and complementary data sources. This technical framework provides essential context for analyzing super-resolution technology integration potential, establishing a foundation for subsequent technical discussions throughout this report.

Annex 3. Implementation Frameworks: Strategic Analysis and Technical Options

This annex provides a comprehensive assessment of implementation pathways for AI-SR technology within CAP monitoring systems. It documents the strategic implications, resource requirements, and organisational considerations associated with both open-source and commercial implementation options. This analysis enables paying agencies to evaluate implementation approaches against their specific technical capabilities, resource constraints, and operational requirements.

Strategic Decision Framework for Implementation

The integration of AI-based super-resolution technology within CAP monitoring systems involves evaluation of both strategic implications and implementation pathways. Implementation represents an organisational commitment requiring assessment of institutional capabilities and resource requirements.

Successful implementation necessitates developing specialised expertise for managing AI-enhanced imagery and associated quality control protocols discussed in Section 3.2 on performance boundaries and scaling constraints, while maintaining compliance with regulatory standards detailed in Section 5. Resource requirements extend beyond initial acquisition costs to encompass continuous quality control processes and staff training, which must be weighed against operational efficiencies such as reduced dependence on traditional VHR imagery.

Paying agencies may consider two distinct implementation pathways: open-source frameworks or commercial solutions. Open-source implementation offers greater control and customization potential but demands substantial internal technical expertise for development, testing, and maintenance. Conversely, commercial solutions provide more immediate implementation potential with reduced technical overhead, though potentially limiting customization options and future adaptability.

The selection between these pathways should emerge from analysis of technical capabilities, resource availability, and operational requirements. This includes evaluation of both current expertise and potential skill development, alongside consideration of implementation costs and long-term sustainability. The chosen approach should align with both immediate operational capabilities and long-term monitoring objectives while ensuring sustainable implementation within available resources.

Open-Source Solutions: Capabilities and Resource Requirements

This annex documents the technical landscape of open-source frameworks for AI super-resolution, with particular emphasis on implementations relevant to remote sensing applications. It analyses architectural approaches, optimisation strategies, and computational requirements associated with frameworks like ESRGAN, SRCNN, Sen2Res, and SR4RS. This technical assessment enables paying agencies to evaluate open-source implementation options against their specific infrastructure capabilities and operational requirements.

Several frameworks have emerged as particularly relevant to remote sensing applications, each demonstrating distinct architectural approaches and optimization strategies. While general-purpose implementations like ESRGAN (https://github.com/xinntao/ESRGAN) and SRCNN (http://mmlab.ie.cuhk.edu.hk/projects/SRCNN.html) have gained widespread adoption, specialised frameworks targeting remote sensing applications have emerged to address domain-specific challenges. The Sen2Res framework (https://step.esa.int/main/snap-supported-plugins/sen2res/), developed by researchers at the INRIA, specifically addresses Sentinel-2 imagery enhancement. Similarly, the SR4RS framework (https://github.com/remicres/sr4rs), maintained by an international consortium of remote sensing specialists, provides specialized architectures for agricultural feature enhancement (Figure 5), often utilizing enhanced Sentinel-2 imagery as input. One such enhancement model is S2DR3 (https://medium.com/@ya_71389/sentinel-2-deep-resolution-3-0-c71a601a225), which represents a major update designed to upscale all 12 spectral bands of a single Sentinel-2 L2A (or L1C) scene from the original 10, 20, and 60 m/px spatial resolution to a target 1 m/px. This model is specifically optimized to preserve subtle spectral variations of soil and vegetation across all 12 multi-spectral bands of Sentinel-2 L2A and can reconstruct objects and textures with individual spatial features.

Figure 5. Comparison of native Sentinel-2 imagery (left) and enhanced imagery (right) processed using SR4RS open-source framework trained on pansharpened Spot-6/7 data.

Source: © *2025*.SR4RS

These open-source implementations typically involve substantial technical infrastructure considerations, where computational requirements scale exponentially with the enhancement factors outlined in Section 2.3. The relationship between model complexity and computational demands influences implementation planning, particularly regarding memory allocation and processing capacity.

The processing pipeline typically involves access to high-resolution satellite imagery or orthophotography for model training, alongside substantial computational resources. While cloud computing platforms have made these resources more accessible, implementation typically involves considerable technical complexity. Paying agencies considering open-source solutions may wish to evaluate their capacity for developing expertise in deep learning, remote sensing, and large-scale data processing.

Commercial Solutions: Service Models and Integration Approaches

This annex examines the commercial landscape for AI super-resolution technology within agricultural monitoring contexts. It documents service architectures, integration capabilities, and operational considerations associated with turnkey commercial solutions. This analysis provides paying agencies with essential information for evaluating commercial implementation pathways, particularly regarding reduced implementation complexity and technical expertise requirements compared to open-source alternatives.

Building upon the foundation established by open-source developments, the commercial landscape for super-resolution technology offers operational alternatives that translate theoretical capabilities into practical monitoring tools. This evolution from research frameworks to commercial products represents a significant technological maturation characterized by architectural adaptations that balance enhancement capabilities with operational reliability.

Figure 6. Comparison of native Sentinel-2 imagery (left) and enhanced imagery (right) processed by a commercial AI-SR service.

Source: © 2025 DigiFarm AS - All Rights Reserved

These commercial implementations leverage deep neural network architectures to achieve automated field boundary delineation, delivering improved spatial resolution compared to native Sentinel-2 imagery. In contrast to open-source alternatives requiring specialised technical expertise, commercial providers offer turnkey solutions that significantly reduce implementation complexity for paying agencies, encompassing the complete processing chain from data acquisition to analysis-ready outputs. However, their practical utility requires systematic evaluation within the context of the limitations discussed in Section 3.2 on performance boundaries and scaling constraints, particularly regarding the risk of generated features that appear contextually plausible but do not reflect ground reality.

Commercial platforms implement multilayer service architectures:

— Cloud-based processing infrastructure handles large-scale satellite imagery enhancement operations.

- Specialised agricultural monitoring modules such as parcel segmentations.
- Standardised API interfaces facilitate integration with existing monitoring systems.
- This approach promotes operational efficiency while requiring an enhanced verification framework to ensure compliance with regulatory requirements for evidence reliability.

Annex 4. Certification and Validation Framework: Technical Specifications for Regulatory Compliance

This annex provides comprehensive technical specifications for the artifact detection and validation methodology outlined in Section 6.3 on artifact detection and validation methodology. It documents the implementation details of metric-based approaches, aggregation frameworks, and visual analytics systems necessary for establishing the reliability of AI-enhanced imagery within regulatory contexts. This detailed technical framework enables paying agencies to implement structured validation protocols that systematically address the challenges of artifact detection while maintaining regulatory compliance.

Metric-Based Detection Approaches for Certification Requirements

Purpose and Scope: This annex provides technical specifications for metrics designed to detect AI-induced artifacts in enhanced satellite imagery. It documents the mathematical foundations, implementation parameters, and detection capabilities of structural, perceptual, and statistical metrics. This analysis establishes the technical foundation for the artifact detection and validation methodology outlined in Section 6.3 on artifact detection and validation methodology, enabling systematic identification of hallucination and memorisation effects in AI-enhanced imagery.

The challenge of artifact identification is further compounded by the sophisticated nature of hallucination and memorization effects in modern SR systems. Advanced models generate contextually coherent artifacts that match surrounding landscape patterns, making them virtually indistinguishable from authentic features during casual examination. This inherent limitation of human visual perception necessitates systematic approaches that can guide expert analysis toward regions requiring verification, rather than relying on exhaustive manual inspection that would prove both inefficient and potentially unreliable.

Addressing this challenge requires specialised metrics designed to detect anomalies that might escape human perception. Wang et al. (2020) provide a review of image super-resolution metrics, categorising them into classical approaches (e.g., MSE, PSNR, SSIM) and advanced perceptual metrics (e.g., LPIPS, MS-SSIM). These metrics evaluate different aspects of image quality with varying sensitivities to specific AI-induced artifacts, creating a complementary analytical framework.

Structural metrics (e.g., SSIM, MS-SSIM) assess how well spatial relationships and patterns are preserved in enhanced imagery. These metrics are relevant for detecting hallucination artifacts that affect landscape feature geometry, field boundary delineation, and spatial relationships between landscape elements. Their sensitivity to structural inconsistencies enables identification of contextually plausible but non-existent features that AI enhancement might introduce in agricultural landscapes. Practical application of these metrics is illustrated through the photovoltaic infrastructure example in Section 3.1 (Figure 2), where geometric regularity metrics would reveal the systematic transformation of linear panel arrangements into organic agricultural patterns. In this case, structural similarity indices would quantifiably demonstrate the degradation of geometric coherence, with regular grid patterns exhibiting significantly lower SSIM scores when compared against reference imagery.

Perceptual metrics (e.g., LPIPS) employ feature representations from deep neural networks to evaluate similarity in ways that better align with human visual perception. These metrics demonstrate capability in detecting subtle hallucination effects that might appear visually plausible while deviating from actual ground conditions. Their particular strength lies in identifying complex artifacts that maintain contextual coherence but misrepresent actual landscape elements.

Statistical and spectral metrics examine distributional properties and frequency characteristics of images. These approaches are particularly valuable for detecting memorisation artifacts, where AI models reproduce patterns from training data rather than faithfully enhancing actual features. Their ability to identify statistical anomalies and unnatural repetition makes them effective for detecting training bias manifestations in enhanced imagery.

A combination of these metric categories provides a valuable detection capability for the artifacts outlined in Section 3.2 on performance boundaries and scaling constraints. The integration of structural, perceptual, and statistical approaches enables systematic identification of contextually coherent artifacts that might otherwise evade detection during visual inspection.

Image quality assessment for super-resolution is a rapidly evolving field. Paying agencies implementing validation frameworks should consider consulting more recent publications to identify state-of-the-art metrics that may have been developed since the time of this report's publication and should select appropriate metrics based on their computational resources and availability of suitable reference images. A balanced approach might incorporate metrics from multiple categories to ensure effective detection capability across different artifact types while maintaining operational efficiency. These metrics serve as technical tools that augment - rather than replace - expert knowledge, providing quantitative assessment that directs specialist attention toward regions requiring detailed evaluation.

The framework's modular design should ensure adaptability to future developments in both super-resolution technology and validation methodologies. As new metrics demonstrating superior performance in detecting specific types of AI-induced artifacts become available, they can be incorporated into the validation framework without disrupting the overall architecture, ensuring the system remains effective even as the technological landscape evolves.

Detection Challenges and Methodological Solutions in Regulatory Contexts

The identification of artifacts in AI-enhanced imagery presents a significant technical challenge due to the fundamental nature of AI super-resolution technology. As discussed in Section 2.1.1 on technical reality versus common perception, contemporary AI-SR architectures are specifically designed to generate photorealistic details that appear convincing to human perception but may not represent actual ground conditions. This creates a complex validation problem: the visually plausible nature of AI-generated artifacts makes them exceedingly difficult to identify through conventional visual inspection, particularly when operating at scale across vast and heterogeneous agricultural landscapes.

This technological challenge directly impacts the certification requirements established in Section 6.2, as the reliability of enhanced imagery within regulatory contexts depends fundamentally on distinguishing between authentic features and AI-generated artifacts. Without systematic validation protocols, the evidentiary value of enhanced imagery for regulatory purposes cannot be maintained, undermining its utility within CAP monitoring frameworks.

The challenge of artifact identification is further compounded by the sophisticated nature of hallucination and memorization effects in modern SR systems. Advanced models generate contextually coherent artifacts that match surrounding landscape patterns, making them virtually indistinguishable from authentic features during casual examination. This inherent limitation of human visual perception necessitates systematic approaches that can guide expert analysis toward regions requiring verification, rather than relying on exhaustive manual inspection that would prove both inefficient and potentially unreliable.

Metric Aggregation and Anomaly Prioritization for Certification Workflows

The validation metrics identified can be combined into coherent analytical outputs that guide expert attention toward potentially problematic areas, creating an efficient verification system where human expertise is directed precisely where needed. This aggregation process transforms individual metric outputs into structured information that highlights potential artifacts, establishing a prioritization framework for expert validation.

A heatmap visualization represents an effective approach to synthesize information from multiple metrics (Equation 1):

Equation 1. Metric Aggregation

$$H(x,y) = w_1 \cdot \text{Metric}_1(x,y) + w_2 \cdot \text{Metric}_2(x,y) + \ldots + w_n \cdot \text{Metric}_n(x,y)$$

Where $\text{Metric}_1 \ldots \text{Metric}_n$ represent the selected validation metrics, and $w_1 \ldots w_n$ are weighting factors that determine the relative importance of each metric in the final assessment. These weighting factors serve as calibration parameters that can be adjusted based on validation outcomes and operational requirements.

While the heatmap approach provides a straightforward and intuitive integration methodology, paying agencies may consider alternative approaches. A particularly effective alternative involves hierarchical filtering systems that progressively refine analysis through sequential application of metrics. This approach first applies computationally efficient metrics to identify regions of potential concern, then applies more sophisticated metrics only to these candidate regions. Such hierarchical systems optimize computational resources while maintaining detection capabilities, making them particularly suitable for large-scale monitoring operations.

Regardless of the chosen integration methodology, the framework should implement a continuous improvement mechanism where expert validation outcomes provide feedback for adjusting integration parameters. This adaptive process enables the system to progressively refine its detection capabilities by identifying configurations that most effectively detect the artifacts addressed in Section 3.2 on performance boundaries and scaling constraints while minimizing false positives. Such adaptability is particularly valuable given the evolving nature of both AI-based super-resolution technologies and validation metrics, allowing the system to incorporate new metrics without fundamental restructuring.

The integrated analysis system should establish a multi-tiered review framework that prioritizes expert attention based on artifact probability. The implementation relies on standardized statistical thresholds that translate metric values into distinct verification requirements:

- Standard monitoring areas: Regions where metrics indicate low probability of artifacts (below statistical thresholds) remain within normal monitoring protocols.

- Enhanced verification zones: Areas where metrics exceed standard deviation thresholds (typically +1 to +2σ from the mean) are highlighted for expert examination during routine quality control processes.

- Critical investigation regions: Locations where metrics show significant deviation (typically beyond +2σ from the mean) trigger immediate and detailed investigation protocols, potentially escalating to multi-source verification using VHR reference imagery or other authoritative data sources.

This tiered approach ensures that expert analysis resources are allocated efficiently, focusing detailed examination on areas with higher probability of containing the hallucination or memorization effects described in Section 2.2 on technological evolution and deep learning approaches. The system recognizes that while computational metrics provide valuable guidance, the final determination of artifact presence requires expert judgment informed by both quantitative indicators and domain knowledge of agricultural landscapes.

The integration framework systematically transforms complex, multi-dimensional metric data into structured information optimized for human interpretation through the visual analytics interfaces described in the next. This transformation enables efficient expert evaluation by directing attention toward potentially problematic regions while providing the necessary context for informed decision-making. The integration process fundamentally recognizes the irreplaceable value of human expertise in artifact validation-computational metrics and their integration provide tools to augment and focus expert analysis rather than automate determination.

The transition from metric computation to integration and subsequently to visual analytics represents a progressive refinement - from quantitative assessment to structured information to expert interpretation - creating a validation pipeline that leverages both computational efficiency and human expertise at their respective points of optimal application.

Visual Analytics Architecture for Expert Assessment and Certification

Visual Analytics represents a specialised analytical discipline at the intersection of interactive visualisation, data science, and human cognition. It integrates computational data processing with human perceptual capabilities, enabling the analysis of complex phenomena through interactive visual interfaces that augment human reasoning capabilities. This interdisciplinary approach is particularly relevant for artifact assessment in AI-enhanced imagery, where purely automated detection may prove insufficient due to the contextually plausible nature of AI-generated features, while purely manual inspection becomes impractical given the volume and complexity of enhanced imagery. This section provides technical specifications for implementing a Visual Analytics system designed to facilitate expert assessment of potential artifacts in AI-enhanced satellite imagery, operationalising the metric aggregation and anomaly prioritisation approaches presented in section on metric aggregation and anomaly prioritisation to guide expert attention toward areas with higher probability of containing AI-induced artifacts.

Within this context, the implementation of an effective validation system requires a systematic approach to data presentation and analysis. The proposed visual analytics framework should integrate multi-source reference data to enable thorough analysis of prioritised anomaly zones. The reference integration layer should incorporate data sources such as orthophotos, VHR satellite imagery, LiDAR data, and ground level photography where available. Each source contributes distinct validation capabilities: orthophotos provide high spatial detail for feature verification, VHR imagery facilitates temporal change detection, LiDAR data offers three-dimensional structure verification, and ground level photography allows detailed examination of feature characteristics at the terrestrial perspective.

The visual analytics interface should provide comprehensive layer management capabilities (including layer activation/deactivation, reordering, grouping, and attribute-based filtering) and fundamental geospatial navigation functionalities (encompassing coordinate-based positioning, extent definition, and spatial bookmarking). These capabilities enable systematic organisation of reference datasets while maintaining clear analytical relationships between enhanced imagery and validation sources.

The interface should implement continuous zoom functionality supporting multiple resolution levels, smooth panning operations, and precise coordinate-based positioning. These navigation capabilities should be synchronised across all active views through coordinate system integration and view state propagation mechanisms, ensuring that pan/zoom operations and extent adjustments in one view automatically trigger corresponding transformations in parallel views. This synchronised visualisation approach maintains consistent spatial reference frames during comparative analysis.

The system should include display controls such as brightness and contrast adjustment, transparency modification (0-100%), and dynamic band combination selection while maintaining geospatial registration integrity. The system should implement comparative visualisation techniques, including transparent overlay capabilities and side-by-side viewing options with synchronized navigation. These visualisation functionalities enable analysts to conduct detailed examinations of anomaly zones and access to ground-level photography when available.

When LiDAR data are available, the system should incorporate vertical profile visualization capabilities that enable analysts to generate cross-sectional views along user-defined transects from these three-dimensional datasets. This LiDAR-based functionality allows precise verification of landscape feature characteristics through quantitative height measurements, which is particularly valuable for validating features with distinct vertical signatures such as tree alignments and hedgerows that might be subject to hallucination effects in AI-enhanced imagery. The vertical profile tool should support interactive transect positioning and dynamic profile generation, enabling rapid comparison between enhanced imagery representations and reference LiDAR data to confirm the authenticity of detected features.

The system should explicitly implement the multi-tiered review framework established in section on metric aggregation and anomaly prioritisation, with visual indicators clearly differentiating between standard monitoring areas, enhanced verification zones, and critical investigation regions based on their respective statistical thresholds. These visual differentiations should employ standardized color schemes that maintain consistency across monitoring operations, with enhanced verification zones (typically $+1\sigma$ to $+2\sigma$ deviations) rendered in moderate emphasis colors, while critical investigation regions (beyond $+2\sigma$ deviation) employ high-visibility indicators that ensure immediate recognition during analytical operations.

To ensure temporal context, the system should maintain synchronisation across all data sources, implementing clear temporal metadata visualisation to account for acquisition gaps and seasonal variations. A temporal profile display should track feature evolution chronologically, enabling visualisation of vegetation seasonal dynamics and landscape transformations. During reference image examination, the system should automatically highlight corresponding acquisition timestamps within the temporal profile, establishing immediate temporal context. This synchronisation mechanism enables efficient differentiation between natural seasonal variations and potential AI-induced artifacts. The system should implement visualisation modes that highlight statistical and structural anomalies, helping analysts distinguish between authentic landscape evolution and AI-generated features.

The temporal profile interface should ideally provide visualisation of reference data availability across the temporal dimension. The framework should implement annotation capabilities supporting geometric features (points, lines, polygons) and attribute data, enabling documentation of observed artifacts.

The visual analytics framework should implement a systematic feedback loop that operationalizes the continuous improvement mechanism described in section on metric aggregation and anomaly prioritisation. This feedback system should automatically capture expert conclusions regarding prioritised anomalies, including validation determinations (confirmed artifact, false positive, indeterminate) and associated confidence levels. It should correlate expert conclusions with the underlying metric values that triggered prioritisation, identifying patterns in metric performance across different landscape types and temporal conditions. The system should generate optimisation recommendations for the anomaly prioritisation system based on historical validation outcomes, progressively refining the system's ability to prioritise genuine artifacts while reducing false positives.

The described framework serves a dual purpose: it provides structured input for the self-improvement feedback loop of the anomaly prioritisation system by identifying systematic patterns in false positives and omissions, while automatically generating validation reports documenting artifact spatial distribution, characteristic patterns, and supporting evidence with appropriate audit trails.

The visual analytics framework represents a critical operational component within the broader certification methodology established in Section 6 on evidential value and certification framework. By enabling systematic detection and documentation of potential artifacts, the system directly supports the fundamental certification need of establishing enhanced imagery's reliability for regulatory purposes. Through its structured approach to anomaly validation, the framework provides the evidential foundation necessary for certification decisions, contributing directly to the relevance of AI-enhanced imagery within the CAP monitoring context.

This visual analytics architecture establishes a methodological foundation for systematic validation of AI-enhanced satellite imagery. The framework's design methodology emphasises analytical rigor through spatial and temporal context integrity, while optimising operational efficiency through automated logging and reporting mechanisms.

While this section provides technical specifications for implementation, it should be emphasised that paying agencies maintain complete autonomy in their technical implementation approaches. The methodological framework presented here should be adapted to specific operational contexts and technical infrastructures, with paying agencies determining the most appropriate implementation pathway based on their particular needs and existing technical capabilities.

While not essential to the core validation methodology, several supplementary capabilities may enhance operational efficiency. These include multi-user collaboration tools enabling synchronized analysis sessions where multiple experts can simultaneously examine regions of interest; data export capabilities supporting standard geospatial formats (GeoTIFF, Shapefile, GeoJSON) with complete metadata transfer and automated report generation; interactive training modules to guide new users through the validation process; and performance optimization through caching mechanisms for frequently accessed reference data.

Reference Data Integration: Technical Architecture for Validation Certification

The visual analytics tools benefit from the integration of multi-source reference data through standardized access protocols, as described below.

This section examines the technical landscape of reference data sources, documenting representative examples of datasets, and establishes foundational protocols for their deployment within the visual analytics environment described in the section on visual analytics architecture.

The Copernicus Data Space Ecosystem (CDSE) exemplifies contemporary Earth observation data infrastructure implementation. This platform enables access to Sentinel constellation data complemented by Contributing Missions offering enhanced spatial resolution. The ecosystem implements three primary access mechanisms: direct data retrieval through the Open Data Hub, programmatic integration via standardized APIs, and Analysis-Ready Data Cube services.

National mapping agencies provide essential reference datasets through high-resolution orthophotos and LiDAR data collections. The French National Institute of Geographic and Information (IGN), for instance, maintains 20cm resolution orthophoto coverage (BD ORTHO®) and comprehensive LiDAR data. Similar capabilities exist through other national agencies like the German Federal Agency for Cartography and Geodesy (BKG) and Spanish National Geographic Institute (IGN Spain), typically implementing standardized WMS and WCS access protocols with free. These institutions typically implement standardized access through Web Map Services (WMS) and Web Coverage Services (WCS).

Ground-level imagery offers complementary capabilities through several systems. The LUCAS database (Land Use/Cover Area Frame Survey) provides systematic ground photography across European agricultural landscapes, implementing a 2 km grid system covering approximately 1.1 million points with triennial surveys from 2006 to 2018. Platforms like Mapillary () complement this coverage through crowdsourced imagery particularly relevant for rural areas, while commercial services like Google Street View provide additional perspectives along transportation networks, implementing usage-based pricing around €7 per 1000 image requests.

Commercial Very High Resolution (VHR) satellite services provide supplementary capabilities when freely available imagery proves insufficient. Planet Labs maintains 50cm-resolution imagery archives accessible through REST APIs. Maxar's SecureWatch platform offers 30-50cm resolution WorldView imagery with viewing-only access models for historical data. Airbus's OneAtlas platform provides similar capabilities through SPOT and Pléiades constellations (50cm-1.5m resolution). These services implement varying pricing structures, with reduced pricing for archive access typically ranging from €2-5 per km^2. The technical implementation of these services generally includes comprehensive API documentation, authentication protocols, and usage monitoring capabilities, enabling systematic integration within validation frameworks while maintaining precise cost control.

The integration framework implements three architectural layers: an access layer managing authentication and usage monitoring, a harmonization layer ensuring coordinate system standardization, and a deployment layer enabling systematic integration through cache optimization and token management. This architecture emphasizes efficient utilization of freely available resources while maintaining clear protocols for commercial data access when required.

Annex 5. Emerging Validation Technologies: Research Trajectories and Implementation Timelines

This annex documents emerging research directions in validation and certification methodologies for AI-enhanced imagery. It provides technical specifications for advanced detection systems, physics-informed neural networks, and temporal consistency frameworks that may eventually strengthen certification capabilities. While these approaches remain in research stages with implementation timelines beyond 2026-2027, this analysis establishes the foundation for future enhancements to the certification framework outlined in Section 6 on legal validity and certification framework.

While paying agencies should base implementation decisions on current technological capabilities and limitations (Sections 2-3 on technical foundations and fundamental constraints), ongoing research suggests potential advancements in certification and validation methodologies relevant to CAP monitoring. This section examines emerging approaches specifically addressing the regulatory and validation challenges identified in this report.

Research in automated artifact detection demonstrates progress toward more reliable validation frameworks for AI-enhanced imagery. Wang et al. (2023) have developed self-supervised anomaly detection systems specifically designed to identify subtle AI-induced artifacts in agricultural landscapes. These approaches show particular promise for detecting contextually plausible but factually incorrect features that might impact payment calculations or compliance verification. Initial benchmarks indicate 15-20% improvement in detection sensitivity compared to current methodologies, potentially strengthening the certification framework outlined in Section 6 on evidential value and certification framework.

Advances in Physics-Informed Neural Networks (PINNs) may partially address the hallucination and memorization effects that fundamentally challenge regulatory evidence standards. By integrating atmospheric scattering models and agricultural phenology knowledge, these systems reduce synthetic feature generation in environmentally complex regions (Cuomo et al., 2022). This approach shows particular relevance for improving reliability in landscape feature detection and parcel boundary delineation - two critical areas identified in Section 5 on operational impacts of AI-SR technology where artificial features could significantly impact regulatory outcomes.

Temporal consistency frameworks represent another promising direction for validation methodology enhancement. Recent research demonstrates that integrating multi-temporal analysis within the validation framework can identify inconsistencies in feature evolution that indicate potential AI artifacts. These approaches leverage Sentinel-2's frequent revisit capabilities (discussed in Section 3.1 on the Sentinel-2 platform) to establish temporal baselines against which enhanced imagery can be systematically evaluated, with potential implementation within the visual analytics framework detailed in Annex 4.

These emerging approaches may eventually strengthen the certification framework outlined in Section 6 on Evidential value and certification framework, particularly regarding artifact detection methodologies and validation protocols. However, paying agencies should recognize that these developments remain in research stages with operational implementation timelines extending beyond 2026-2027. Current implementation decisions should therefore prioritize the established certification protocols and validation methodologies detailed in Section 6 on Evidential value and certification framework, which provide a framework for managing current technological limitations.

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